



C & I MAINT Department
SIMHADRI

Note No :1



Ref:SMTTP/C & I MAINT/2025-26/CIHMI/270020

Date:26/06/2025(19:03:18)

Subject:approval for implementation of LMI on Alarm Management System

Alarm optimization is an iterative process consisting of:

1. Data collection, data analysis
2. Fixing physical problems
3. Identifying unnecessary alarms and eliminating them
4. Making enhancements to required alarms.
5. Creating and or updating documentation
6. Performing management of change (MOC) along with performance review
7. Implementing, testing, and repeating the above process periodical manner

Based on OGN ref no:**COS-ISO-00-OGN/OPS/C&I/005 REV-00** on “Guidelines on Alarm Management System ,LMI on Alarm Management System has been prepared for NTPC Simhadri.

The objective of this LMI is to ensure that necessary alarms are annunciated which have pre-defined operator actions with suitable priorities configured in alarm system for operator to effectively manage the plant or process upsets.

Prepared LMI and OGN :COS-ISO-00-OGN/OPS/C&I/005 has been attached for reference.

Put up to competent authority for review and approval for implementation of LMI on Alarm Mangement system.

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Approval may kindly be accorded.

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Note No:3

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Note No:5

It is requested to modify as per standard format of Simhadri LMI (Guidelines for Issue of LMIs as a Controlled Document) LMI/OGN/GEN/002 Rev:04 as uploaded on Simhadri EEMG LAN

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Note No:6

Required Changes incorporated as per standard format of LMI and revised document attached.

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Note No:10

LMI is in line with the OGN. May please be approved.

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LMI is in line with OGN as in Note #13.

Submitted for kind approval, please.

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NTPC

SIMHADRI

LOCATION MANAGEMENT INSTRUCTION

ALARM MANAGEMENT SYSTEM

LMI/OGN/OPS/C&I/005 Rev.: 00

JUNE-2025

Approved for Implementation by HOP (SIMHADRI)

LMI APPROVAL DOCUMENT

1	LMI NAME	Alarm Management System
2	LMI No.	LMI/OGN/OPS/C&I/005 Rev.: 00
3	Reference Document	COS-ISO-00-OGN/OPS/C&I/005 REV-00
4	Superseded Document	NIL
5	Prepared By	Ajeet Kumar Raushan, Sr.Manager (C&I)
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TITLE	Alarm Management System
LMI NO:	LMI/OGN/OPS/C&I/005 Rev.: 00

1.0 Introduction

1.1. Purpose and functions of LVS (Large Video Screens) alarms are,

- 1.1.1. To direct the operator's attention towards the problems, abnormal conditions or process changes that need timely assessment and/or timely corrective action(s) to bring the plant back to normal conditions.
- 1.1.2. To provide feedback on the corrective actions taken by the operator in response to the alarms. (Supplementary alarms)
- 1.1.3. To bring the notice of operator about Un-safe operating, Hazardous or deviation from normal or desired conditions before Emergency Shutdown of plant or equipment is triggered.

1.2. In the 1970s, there were approximately 60 to 100 alarms configured per operator. Each alarm was designed to be highly reliable and available. Due to limitations in component design, they were bulky, relatively costly, and highly proprietary in design. This resulted in an overall alarm system that was manageable with respect to quantity of alarms; however, the system was not very flexible.

1.3. Introduction of DCS HMI/SCADA systems facilitated to create alarm systems that are flexible, small, and relatively inexpensive. However, Modern alarm systems can generate more than 1,000 alarms per operator per day, if unoptimized. This overload effect can create several undesirable outcomes, including:

- a. Causing a process disturbance leading to trip or to last longer than necessary.
- b. Causing a catastrophic equipment, system and/or plant failure, as happened at in many process plants worldwide, where protections failed to act or not envisaged
- c. Creating increased stress for operators that can lead to poor judgment and decreased morale.

- 1.4. Resolution to alarm system problem is multi-disciplinary. Alarm optimization is an iterative process consisting of:
- a. data collection, data analysis
 - b. fixing physical problems
 - c. identifying unnecessary alarms and eliminating them
 - d. making enhancements to required alarms.
 - e. creating and or updating documentation
 - f. performing management of change (MOC) along with performance review
 - g. implementing, testing, and repeating the above process periodical manner

The objective of this LMI is to ensure that necessary alarms are annunciated which have pre-defined operator actions with suitable priorities configured in alarm system for operator to effectively manage the plant or process upsets.

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2.0 Records of revision

Nil

3.0 Roles and Responsibilities

3.1 Management shall demonstrate commitment with respect to alarm management system by

- 3.1.1 Constituting a team for each stage or offsite areas consisting of two persons from operation and one each from C&I, Electrical and EEMG as alarm optimization team with min 5 years of experience.
- 3.1.2 Reviewing the alarm optimization team on annual basis.
- 3.1.3 Providing adequate resources for the team.

3.2 The team member from C&I & Electrical shall be responsible for their respective area for collecting the alarm data for given period and analyze the data for chattering, frequent, stale, and unnecessary alarms as per the alarm philosophy guide. Generation of quarterly performance reports for operations and maintenance of alarm system. Ensure timely maintenance of sensors and instruments causing nuisance alarms. He shall Ensure time synchronization of DCS (Distributed control system) systems with GPS clock (Multiple DCS cases). Ensuring availability of alarm and SOE (Sequence of Events) system prior to P&I checking during recommissioning post overhaul.

3.3 The team member from operation shall be responsible for preparation of alarm operator guidance document. Provide alarm setpoints and change request for any new alarm or change in alarm. Notification of nuisance alarm and chattering alarms to respective C&I/Electrical maintenance. They shall verify of alarm database w.r.t respect to Annexure 4. The shall attach the print of alarm and SOE log in P&I records as per circular dated 18.03.2016(ref).

3.4 The team member from EEMG shall be responsible for coordination of Team meeting alarm management and document the results on annual basis. Develop baseline report for performance of alarm system in the unit. Member from EEMG shall be responsible for change management of alarm optimization team in case of members non-availability at stations.

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4.0 Guideline for Alarm Management System

The following are the instructions shall be followed for DCS system wise management of alarm system.

4.1 Develop, Adopt & Maintain Alarm Philosophy

The first step in alarm management is that all stations shall develop, adopt and maintain an alarm philosophy document. Key points for Maintaining and Designing an Effective Alarm System are,

4.1.1 Presenting only relevant and useful alarms to the operator

4.1.2 Each alarm should have a defined response from the operator.

4.1.3 Allow adequate time for the operator to respond to an alarm.

4.1.4 Configure and present only good alarms:

4.1.4.1 Alarms that help an operator in monitoring and controlling the plant/equipment.

4.1.4.2 Alarms should not be a nuisance or hindrance for the operator.

4.1.4.3 Operator's time and attention should not be diverted on the alarms which did not require any response or intervention from the operator and can be ignored. Otherwise, there is a possibility of the "Cry Wolf" effect to ignore alarms.

Annexure-1 to this document contains design philosophy for new alarm addition and rationalization of alarms. This document shall be used for deciding alarm priority, setpoint selection. etc. of new alarms and management of the existing alarms.

4.2 Collect Alarm data and benchmark the system.

The alarm data (monthly) shall be collected and analyzed in excel or alarm analysis software (if available) or power BI. From the monthly data (excluding operator action),

- a. Trending of an alarms per day graph
- b. Top 10 annunciated alarms
- c. Alarm priority distribution
- d. Chattering alarms, alarms in 10 minutes (>10 alarms in 10 minutes as flood)
- e. Stale alarms (Long standing) and
- f. Alarm configuration analysis shall be performed

Based on the collected data, An initial alarm system baseline and benchmark against industry best Practices shall be performed to planning the improvement process.

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4.3 Perform “Bad Actor” Alarm Resolution

Based on the analysis of hundreds of systems, there are several varieties of nuisance or “bad actor” alarms. A relatively small number of tags produce large numbers of alarm events. Identify and address the existing nuisance alarms (chattering, stale, and duplicate alarms) on the system. With a concerted effort many of the nuisance alarms in the system can be addressed immediately. The average improvement is over a 50% reduction in overall alarm events for a relatively minimal effort. While on some systems this result may not meet an overall improvement goal, it is a great first step, providing much needed immediate relief.

4.4 Process wise alarm distribution (Main Plant)

Alarms may be configured and distributed as per the following. Where two tier LVS envisaged, it shall be in bottom row LVSs. The number of LVS is starting from left facing the LVS. Alarm hooters or audio visual shall be enabled for Low (bottom band), high (middle band), and emergency (top band) priority alarms.

- a) LVS01 - SG Integral process alarms along with its system alarms
- b) LVS02 - SG Aux /SGBOP process alarms along with its system alarms
- c) LVS03 - TG Aux/BOP process alarms along with its system alarm
- d) LVS04 - TG integral alarms along with its system alarms.
- e) LVS05 - Electrical system alarms along with its system alarm

4.5 Perform Alarm rationalization and Documentation (master database)

5.3.1 Effective techniques for improving alarm systems: Often, most of the alarms can be improved by carrying out simple modifications. Various techniques used in practice and the benefits derived from these techniques are classified as follows:

- a. Techniques that are effective and result in high benefits are:
 - i. Review alarm behaviour after all disturbed/upset conditions to confirm usability.
 - ii. Adjust dead bands of repeating/nuisance alarms, Fine tune alarm settings if required.
 - iii. Eliminate alarms that require no operator response/corrective action(s).
 - iv. Ensure that critical and high priorities are allocated only to appropriate alarms related to safety, economic/generation loss, and environmental consequences.
 - v. Review the alarm messages(text) that operators do not understand or are unaware of the response required.
 - vi. Alarm shelving facility (at discretion of committee).

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- b. Techniques that require more efforts and result in medium benefits are:
 - i. Suppress alarms related to section/sub-section of the equipment out-of-service. (at discretion of committee).
 - ii. Use of filters, transient suppression, and de-bounce timers for repeating alarms (only for alarm part of the logic).
 - iii. Replace digital alarm sensors causing nuisance with analog sensors.
 - iv. Review alarms from bad signals and actuator discrepancy alarms causing nuisance.
 - v. Apply logic to simplify and combine alarms that are redundant by grouping.
 - c. Other techniques, including some of the advanced techniques are:
 - i. Automatic adjustment of alarm settings depending on plant condition.
 - ii. Introduce additional operator set alarms for pre-decided parameters.
 - iii. Grouping of alarms that require the same operator response with facility for navigation to specific alarm.
 - iv. Based on mode of operation of the plant, automatic suppressing of alarms.
- 4.5.1 Every DCS/PLC system shall have alarm database of configured alarm.
- 4.5.2 The rationalised alarm database shall be prepared after achieving benchmark performance of alarm system. Alarm database shall have setpoints, priority (Band), LVS (Large Video Screens) number and tag number for each alarm point.
- 4.5.3 Alarm rationalisation is often misunderstood as deletion of alarms. Rationalisation is acting upon the frequent alarms that are occurring and acting upon them as per the following tips.
- 4.5.3.1 Eliminate false alarms with action plan for restoration, remove redundant alarms, masking of alarms, out of service equipment alarms.
 - 4.5.3.2 Operators veto power: Many operation engineers argue for “good to know” alarms persisting in alarm facia. But thumb rule says that “If there is no specified action for the operator to do on an alarm, there should not be an alarm in the first place”.
 - 4.5.3.3 Addressing to 10 “Bad actor” Alarms; Usually very few alarms cause half of the entire alarm load. Hence, addressing the bad actors will improve the alarm system performance.

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The following table has common problems with probable causes.

S No	Alarm management problem	Cause(s)
1	Alarms are generated but they are ignored by operator	Nuisance alarms (Chattering and fleeting alarms) Hardware faults, redundant alarms, cascade alarms, Incorrect alarm settings and alarms not rationalised.
2	When alarm occur, operator do not know how to respond	Insufficient or no alarm response guide; good to know alarms; lack of training.
3	Minor unit emergency cause flooding of alarms	Average alarm load is too high, redundant alarms, cascade alarms, Incorrect alarm settings and alarms not rationalised
4	The alarm facia is full of alarms; even though normal operation	Good to know alarm, Nuisance alarms (Chattering and fleeting alarms) Hardware faults, redundant alarms, cascade alarms, Incorrect alarm settings and alarms not rationalised
5	Some alarms are present in facia more than 24 hours.	Good to know alarms, Corrective action is ineffective. Equipment or instrument faulty
6	Alarm settings are changed from one operation group to the other.	Lack of change management of change procedures.

4.5.4 Alarms database is prepared from following documents, the same shall be easily accessible by the rationalisation team.

- a) P&IDs
- b) Approved DCS Logic documents/ DCS Configuration data / Network Alarms
- c) OS- Circulars/Communications/OGNs/ODs/OINs
- d) Trip committee recommendations
- e) Safety committee or HAZOP (Hazard Operability) study reports.
- f) UPS/CHARGER/TSI/Flame scanner manuals
- g) DCS trends/PI trends for suitable setpoint decision.
- h) Modifications implemented at stations (change request records)

4.5.5 Every unit DCS wise alarm master database after rationalisation shall be updated.

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4.6 All the analog signals used for alarms (LVS annunciation) shall be configured in historical trend. Alarm Testing and change management

- 4.6.1 It is important ensure that the alarm system is working reliably. There shall be annunciation checking provision to know that the alarm system is working (auto generated at 8 hours preferably).
- 4.6.2 All the alarms shall be tested during initial commissioning (including R&M) of the unit.
- 4.6.3 The alarms related to equipment protection and hazardous process shall be checked during P&I checking of the equipment. Other important alarms shall be checked during overhauling / recommissioning if they are justified and viable for testing. Remaining alarms shall be verified in the alarm database.
- 4.6.4 To maintain the integrity of the facility, the Annexure-2 format for change management shall be used to control and document all modifications to the alarm system. Procedure for modification of highly managed alarms (equipment protection or safety systems and hazardous process) shall be as per OD-GEN-006 (System for plant modifications).

4.7 Alarm System Performance assessment & Reporting

- 4.7.1 For assessment of the plant alarm system performance, operator feedback shall be taken as per the reference questionnaire every two years.
- 4.7.2 Report of DCS wise most frequent alarms with tag name shall be generated quarterly. The tags generating most alarms shall be resolved through optimisation committee.
- 4.7.3 After alarm rationalisation a base line performance report shall be prepared and the same shall be compared with quarterly performance or current performance of the alarm system. Performance of alarm system shall be reviewed/discussed in DPM every quarter by the station management.
- 4.7.4 Alarms persisting for more than 24 hours shall be communicated or notified to the concerned maintenance department. If the alarm is due to process or equipment constraint, increasing the alarm setpoint suitably to be explored and noted down in the protection bypass register along with expected normalisation date so that the equipment pre trip warning is not missed.
- 4.7.5 Alarm management system audit shall be conducted as per the audit frequency of technical audit. (Ref COS-ISO-00-OIN/GEN/004)

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The target alarm system performance shall be as per the following table. An interim target may be kept at double of the final target for an initial one year. The targets provided in the following table are per LVS/Console/operator. Alarm System performance should be periodically reported by the Alarm Management System committee. The report should be distributed to Heads of O&M, Operation, and shift in-charges.

Analysis Type	Metric	Final Target
Configuration Information	Frequency distribution of configured alarms.	Emergency: below 5% High: below 15% Low: above 80%
Dynamic alarms	Long term average alarms/ day	<288 / day
Dynamic alarms	Peak alarms/ 10 min	<20 / 10 min
Dynamic alarms	Standing alarms	<10 at any time
Dynamic alarms	Disabled (shelved) alarms	<30 at any time
Dynamic alarms	Chattering alarms (activated 5 times or more per minute)	0
Dynamic alarms	Stale alarms (in alarm state longer than 24 hours)	0
Dynamic alarms	Risk time – long term	0
Dynamic alarms	Risk time – peak	0

In addition to the KPIs above, the report should also include:

- a. List of alarms which are not in service (e.g., shelved, bypassed)
- b. Frequency analysis for most frequent alarms and chattering alarms, showing top 10
- c. List of long standing (stale) alarms
- d. Creation of a Nuisance Alarm List and progress made against the resolution of the alarms on that list
- e. Definition of action plans to improve performance compared to the KPIs, and progress of those plans

4.8 Alarm response procedure for operation (guide/manual)

- 4.8.1 Each alarm shall have a defined operator action response, which shall be in the form of a Preventive and/or Corrective Action(s) or an Acknowledgement.
- 4.8.2 For ease of access and maintainability, the alarm system documentation shall be maintained through a uniform electronic database system on intranet and hard copy in control room. At a minimum, the following items shall be documented for each alarm, either during initial alarm design or during rationalization.

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- a) Potential consequences if the operator does not respond to the alarm.
 - b) Maximum time available for the operator to respond and mitigate identified consequences.
 - c) The reasons for over-riding any priority recommendations.
 - d) Operator response or corrective actions for the alarm
 - e) Causes of the alarm.
- 4.8.3 Alarm response operator guide / manual shall be prepared by operation Engineers as per the format attached in Annexure-3 for each alarm points. During preparation of the manual respective process or equipment owners (Boiler, Turbine, C&I and Electrical) inputs may be taken.

5.0 Records to be maintained.

S. No.	Record Description	Maintained by	Frequency	Remarks
1	List of Alarms (Alarm database)	Head of C&I	Continuous	As per Annexure 1
2	Alarm response manual / guide for operator	Head of Operation (Hard copies)	Continuous	As per Annexure-5
3	Quarterly Alarm performance report.	C&I / Electrical	Quarterly	
4	Change management records	Head of C&I		As per Annexure 2
5	Alarm rationalization committee records & Minutes	EEMG	Annual	(Quarterly during initial one Year)
6	Alarm feedback from operators	Committee	2 yearly	Annexure 7 before every overhaul.
7	Back up of LVS alarms logs / DCS event logs	Head of C&I	As per OGN-OPS-C&I-004.	

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6.0 **Review:** HOD (C&I) will review & take the approval of competent authority (HOP).

The document will be reviewed on 2 yearly basis or earlier, if required.

7.0 **Checklist:** As per annexure.

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8.0 Annexures

Annexure 1: Alarm Master Database Format

S. No	DCS Tag	Alarm Description	Alarm Setpoint	LVS No	Priority	Owner	Ref of Alarm
1							
2							
3							
4							
5							
6							
7							
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26							
27							

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Annexure 2: Alarm change Management Format

Criticality: Emergency High Low	Applicable Units/ Status (1 / 2 / 3 / 4)	Submittal Date:	Completion Date:
Area:	Control System:	Change Type: Addition De-commission Modification	
Change requester:	Department:	i. Highly Managed or Protection or safety Alarm	
Alarm Point detail:			
Description of change request:			
Justification of change request:			
Impact on non-implementation:			
Change Reviewer	Alarm point revision	Alarm change management ID:	
Change Impact analysis: (required time, applicable systems & documents, other costs)			
Implementation plan, required tests, training needs:			
Implementation approver:	Approval date:	Approver's signature:	
	Implementation date:	Implementer's Signature:	

Date

Commissioning complete:

..... Testing

complete:

Alarm database Updated:

..... Documentation

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Annexure 4: Alarm Management System audit checklist

S.No	Clause	Key Requirement	Evidence sought	Responsibility
1	4.1	Compliance to alarm design philosophy	Check change management records for new alarm addition and compliance to 4.1	C&I
2	4.2	Process wise distribution of Alarms	Verify Alarm database for process wise distribution of alarms.	C&I
3	4.3	Alarm master database	Verify availability of documented alarm database as per annexure 1	C&I
4	4.3	Alarm rationalization	Verify the records for alarm rationalization for its effectiveness. Verify availability top quarterly top 10 bad actors and action plan for addressing them	C&I
5	4.4	Alarm response procedure manual / guide for operator	Check for availability of documented alarm response procedure as per annexure 3	Operation
6	4.5	Alarm testing and change management	Check for availability of defined testing frequency for each alarm type. Check for documented protocols for critical alarms like equipment protection etc. Check for change management records for alarm system	C&/Operation / Electrical
7	4.6	Alarm system performance Assessment	Verify availability of documented alarm system performance targets. Check for records of alarm system performance measurement and monitoring. Check for operator feedbacks and improvement of alarm system. Verify periodical review of the alarm management system performance	C&/Electrical
8	5	Alarm optimization committee	Check availability of alarm optimization committee Check record of minutes of meeting of annual performance review of the alarm management system	Committee

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Annexure 5: Alarm Management Philosophy Guide (Reference)

Introduction

The O&M of power plants requires the availability of adequate information by way of alarms, to make possible correct decisions which are essential for maintaining the safety and reliability of the equipment. Process and system alarms are essential to monitor plant conditions and attract the attention of the process plant operator to significant changes that require timely assessment or action. A well configured alarm system helps the operator to correct potentially dangerous situations before the unit emergency shutdown or protection operates, thus improves plant safety, reliability, and availability. Ineffective alarm systems have often been cited as contributing factors in the investigation or trip reports following major incidents and unit outage.

To accomplish the above objectives, each alarm point in the system must:

7.1.1 Require an operator action.

7.1.2 Provide sufficient time for operator action.

7.1.3 Be designed based on a consistent and documented approach.

This guidance note is applicable for assessment / improvement of existing alarms of stations and during design of alarm system for upcoming projects.

Type of Alarms and Alarm Priority:

Alarm Priority: Alarm priorities are used to define the relative importance of an event. Throughout this document, it is assumed that the control system has five (5) alarm priorities: No Alarm / Journal / Low / High/Emergency.

No Alarm: No alarm is configured.

Journal: The alarm is recorded and time-stamped in the control system archive, but no visual or audible alarm is generated to alert the operator.

Emergency/High/Low: These alarms are raised to the operator console and generate an audible alarm. The alarms are also recorded and time-stamped in the control system archive.

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Type of Alarm:

- 7.1.3.1 Bad I/O Alarm: A system alarm that indicates an inaccurate or unavailable input or output
- 7.1.3.2 Absolute Alarm (HH (High High), H, L, LL (Low Low)): A process alarm that an analog signal exceeds a pre-defined abnormal value (the alarm set point). Most control systems have the following absolute alarms:
 - 7.1.3.2.1 High-High (HH)
 - 7.1.3.2.2 High (H)
 - 7.1.3.2.3 Low (L)
 - 7.1.3.2.4 Low-Low (LL)
- 7.1.3.3 Change-of-State Alarm: A process alarm notification that equipment has changed its operation from one state to the other. For example, a pump changes from "On" state to "Off" state, and vice versa.
- 7.1.3.4 Command-Disagree Alarm: A process alarm of an unexpected equipment state compared to its actual state. For example, an MOV status is different than its command state (after a suitable time delay).
- 7.1.3.5 Deviation Alarm: A notification that the difference between the setpoint and the actual PV (Process Variable) exceeds a certain limit.
- 7.1.3.6 Off-Normal Alarm: A notification that equipment is not in its normal state. For example, if the normal state of an MOV is open, then Off-Normal alarm would occur when the MOV is not in its normal, open state.
- 7.1.3.7 Plant-State Alarm: An alarm optimization technique used to change the configuration of one or more alarm points in real-time, depending on the operating states of the equipment.
- 7.1.3.8 Process Alarm: A notification that a changed condition has been detected in the operating facility; either a process condition (e.g., a flow) or a process state (e.g., a pump stopped). How the notification is reported depends upon the priority of the process alarm.
- 7.1.3.9 Rate-of-Change Alarm: A notification that the rate-of-change of an analog process value exceeds a pre-defined abnormal value (alarm set point).
- 7.1.3.10 Control System or instrument Alarm: Alarm which occurs because of DCS hardware or software or instrument is fault.
- 7.1.3.11 Operator set alarm: Operator-set alarms are the alarms for which the alarm settings are adjusted by the operator to suit his needs while operating the plant. Some temporary alarms on process variables and settings chosen by the operators are also included in these alarms. However, it is useful to have the operator set alarms with some password.

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- 7.1.3.12 Adjustable alarms: Adjustable alarms are an advanced form of absolute alarms in which the alarm settings are adjusted to suit the operating conditions. Adjustment of alarm setting is done automatically based on pre-defined/pre-programmed logic.
- 7.1.3.13 Adaptive alarms: Adaptive alarms are advanced alarms generated using deviation alarms or rate of change alarms in combination with the absolute alarm limits. When the parameter is away from the absolute threshold limits, the allowable deviation or rate of change limits for the parameter is relaxed. But when the parameter is near the absolute threshold limits, the limits on deviation or rate of change alarm are tightened to control the parameter within the threshold limits effectively.
- 7.1.3.14 Re-triggering alarms: Re-triggering alarms are automatically re-annunciated to the operator based on certain conditions. One such condition can be when the alarm remains un-acknowledged by an operator for more than a pre-defined time setting. Other conditions for generating a retriggering alarm could be the alarmed variable moving significantly, exceeding its set limits.
- 7.1.3.15 First-up alarms: First-up alarms are not exactly a type of alarm, but rather a facility for identifying the sequence of alarm occurrence. First-up alarm helps in identifying the reason for the plant/equipment trips. In some of the control systems available on the market, the alarm annunciation systems have built-in facilities to display first-up alarms. The number of alarms is put in a first-up alarm group. The first alarm detected in the group is displayed blinking/flashing and the following alarms are displayed as steady on the group alarm display.
- 7.1.3.16 Statistical alarms: Statistical alarms are generated using statistical techniques to filter out significant changes from the process noise.
- 7.1.3.17 Signals that should not be alarms: It is important to note that the following signals should not be alarms; however, these may be displayed in the events list or as a status display on the plant mimic panel or graphics:
- Plant status change or process variables that do not require any operator attention.
 - Signals without a defined operator response.
 - Alarm warnings that are too fast for the operator to respond or take corrective action.
 - Alarms/events that are logged or recorded in Alarm Logs and Events Logs – which the operator does not need to see during operating the plant.
 - Signals which only confirm the successful actions taken by the operator.
 - Signals which are a duplication of another alarm.

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Procedure for addition of new Alarm

The alarm system design will determine if an alarm is required. If an alarm is required, then.

What is the priority setting? What is the alarm setpoint and other configuration settings? Is it appropriate to modify the alarm settings based on the operating state of the equipment? And How alarms are visually and audibly presented to the operator?

The questions below are used to determine conditions when a process alarm is necessary to be raised to the operator (Emergency, High, or Low priority).

Does the event represent an abnormal condition? Does the abnormal situation require operator action? Is there adequate time for an operator action?

Is this the appropriate place to indicate to the operator the event has occurred?

Those events that do not meet the any of the above criteria should be either Journal priority (stored in history) or No Alarm (not configured). One guiding principle is this: if there is no abnormal condition, then no alarm shall be raised. For example, if a pair of 100% pumps are provided, it is a normal condition for one of the pumps to be on and one off. The offline pump should not have a standing alarm. Conversely, there are some pumps that have no running spare. When "running" is always the normal condition and "stopped" is always the abnormal condition, an alarm triggered by transition to stopped may be needed (subject to the remaining criteria).

If an alarm is required, analysis must be conducted to determine the appropriate alarm priority.

Alarm Priorities:

Alarm priority is a means to convey the urgency of a specific process condition to the operator and drives the operator's responses during upsets when many alarms occur simultaneously. To properly focus the operator's responses, a logical and consistent method of assigning priorities is required.

Two crucial factors will be considered when determining the priority of an alarm:

- Severity of Consequences
- Maximum Response Time

There are three levels of alarm priority for the alarm system that are raised to the operator. These levels should be consistent for all DCS and PLC (Programmable Logic Controllers) system. Fire detection system is excluded from this categorization because it is an independent system. EEMUA recommends the following alarm priority distribution as a guideline:

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Alarm Priority	Percentage of Alarms
Emergency	5
High	15
Low	80

A way to assign priorities via the use of matrices is shown below.

Areas of Impact and Severity of Consequences:

The selection of an alarm priority depends heavily on the consequences of the abnormal condition if the operator fails to take corrective actions in a timely fashion. For each alarm, the potential consequences without any operator actions must be identified. A risk matrix is to be defined for each individual plant site. A risk matrix will be used to define the Severity of Consequence criteria. An example risk matrix is shown below.

	Severity of Consequence		
Impact	Minor	Major	Severe
Personnel	None	Lost time injury	Life Threatening
Public or Environment	Minimal exposure. No impact. Does not cross the plant boundary. Contained release. Little, if any, clean up. Source eliminated.	Exposed to hazards that may cause injury. Hospitalizations and medical first aid possible. Damage Claims.	Exposed to life-threatening hazard. Disruption of basic services. Impact involving the community. Catastrophic property damage. Uncontained release of hazardous materials with major environmental impact and 3rd party impact.
Plant/Equipment	Equipment damages that result in negligible unit downtime or partial loss.	Results in unit downtime up to 15 days, some to severe equipment damage.	Results in loss of entire unit or critical equipment for more than 15 days

Maximum Response Time:

Maximum Response Time (MRT) is the time that the operator must take action to prevent or mitigate undesired consequences caused by an abnormal condition. The shorter the time to respond, the higher the priority of the alarm, assuming equal consequences can result. MRT must include the action of outside personnel following direction from the console operator.

MRT categories shall be defined for each individual plant site. MRT will be defined for each alarm. An example MRT table is shown below:

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Maximum Response Time Categories: > 30 minutes, 10 to 30 minutes, 3 to 10 minutes and < 3 minutes.

Alarm Priority Matrix:

Combining the severity of consequence and the MRT provides a systematic approach for setting alarm priorities. An alarm priority matrix shall be defined for each equipment. Each alarm priority is determined using this matrix. An example alarm priority matrix is shown below:

Maximum Response Time	Severity of Consequence		
	Minor	Major	Severe
>30 Minutes	No Alarm	No Alarm	No Alarm
10-30 Minutes	No Alarm	Low	High
3-10 Minutes	Low	High	High
<3 minutes	High	Emergency	Emergency

Alarm Settings:

Alarm settings include alarm setpoints and alarm filter configuration settings. These settings should be selected to provide adequate response time to plant operations and to minimize nuisance alarms.

At design time, alarm setpoints are determined by the process design engineer. Care must be taken to ensure that operations have adequate time to take action to correct the abnormal situation. After sufficient process experience has been gained, the setting shall be adjusted to reduce the occurrence of nuisance alarms.

To minimize chattering alarms, which activate repeatedly over a brief period, appropriate alarm dead bands and digital delay times are recommended. The recommended design settings are shown below. Adjustments to these values may be made after sufficient operating experience has been obtained.

Signal type	On Delay Time (Digital)	Dead Band (Analog)
Flow	2 sec	5%
Level	2 sec	5%
Pressure	1 sec	2%
Temperature	0 sec	1%

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Alarm response operator guide/manual

For ease of access and maintainability, the alarm system documentation should be maintained through a uniform electronic database system on intranet. At a minimum, the following items shall be documented for each alarm, either during initial alarm design or during rationalization.

1. Potential consequences if the operator does not respond to the alarm.
2. Maximum time available for the operator to respond and mitigate identified consequences.
3. The reasons for over-riding any priority recommendations.
4. Operator response or corrective actions for the alarm
5. Causes of the alarm

Alarm response operator guide / manual shall be prepared by operation Engineers as per the format attached in annexure1 for each alarm points. During preparation of the manual respective boiler, turbine, C&I, and electrical department' inputs shall be taken during preparation of the same.

Alarm change management.

To maintain the integrity of the facility, the annexed format for change management shall be used to control and document all modifications to the alarm system. For changing/increasing alarm setpoints of unit protections, OD-GEN-006 procedure for "System for plant modification" shall be used. This procedure includes the review and concurrence by Engineering and OS.

Alarm System Performance Analysis and

Monitoring Baseline Analysis

A "Baseline" consists of a group of analyses that are conducted using both static and dynamic information. The dynamic alarm data should cover six weeks or more to establish a normal, average operating condition. It is recommended to perform a Baseline at all the units where this has not previously been conducted. A Baseline should be used to verify the proper design of alarms after commissioning a new project.

Purposes:

- Document existing condition for later comparison to measure improvement
- Measure alarm system performance (measure risk)
- Pinpoint locations (consoles) of most serious risk
- Determine major problem root causes (too many alarms, incorrect alarm setpoints, faulty instruments, etc.)
- Compare subject consoles to other stations and worldwide targets. Analyse configuration metrics:

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Analysis Type	Metric	Target
Configuration Information	Alarms configured/ recommended maximum alarms configured (recommended max. = $6 \cdot C + 2 \cdot AI + .6 \cdot DI$ where C = no. of controllers, AI = no. of analog inputs, DI (Digital Inputs) = no. of digital inputs)	<1.00
Configuration Information	Frequency distribution of configured alarms	Emergency: at or below 5% High: at or below 15% Low: at or above 80%
Dynamic alarms	Long term average alarms/ day	<288 / day
Dynamic alarms	Peak alarms/ 10 min	<20 / 10 min
Dynamic alarms	Standing alarms	<10 at any time
Dynamic alarms	Disabled (shelved) alarms	<30 at any time
Dynamic alarms	Chattering alarms (activated 5 times or more per minute)	0
Dynamic alarms	Stale alarms (in alarm state longer than 24 hours)	0
Dynamic alarms	Risk time – long term	0
Dynamic alarms	Risk time – peak	0

The configuration of the alarm system requires ongoing maintenance to ensure accuracy, operability, and compliance with plant standards. Once an alarm database is developed, the effective management of that alarm database requires performance assessments and a proven workflow process.

Weekly Bad Actor Report

A weekly report of bad actors enables the alarm optimisation team (Operations, Maintenance planning, EEMG) to focus quickly on the instrument loops that are causing the most alarms in various categories. At the alarm optimisation team level, root cause should be determined, and the problem corrected. This procedure is beneficial to give the operator relief whenever alarm rates are high.

But, where alarm rates are consistently or significantly above recommended limits, the only permanent solution is a comprehensive rationalisation of all the alarms (as per SI.No.4 procedure of addition of new alarm). Weekly reports are also useful for the C&I maintenance to monitor operator actions (e.g., which alarms were disabled during the week), and pinpoint which controllers have required operator changes most frequently.

Controllers needing frequent attention by operators (mode changes, setpoint changes, output changes) indicate problems that need engineering and tuning attention.

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Recommended weekly bad actor reports should initially include the following analyses, which can be adjusted based upon operating experience.

- Average alarm rate
- Chattering alarms (>5 times/minute)
- Stale alarms (persisted>24 hours)
- Duplicate alarms
- Consequential alarms – alarm on an operator action or auto action.
- Alarms disabled/enabled
- Controller mode changes
- Controller setpoint changes
- Controller output changes

Monthly Key Performance Indicators (KPIs):

The following table shows the alarm system Key Performance Indicators (KPIs) for long term monitoring (based on a single operator console).

Key Performance Indicator (KPI)	Monthly Target
Long Term Alarm Rate	0 days exceeding 288 alarms/day
Peak Alarm Rate (defined as the number of alarms for 10-minute time)	0 10-minute slices exceeding 20 alarms/ 10 minutes
Inhibited/Disabled Alarms	< 30 (Unless as part of defined Shelving or State-based Strategy)
Chattering Alarms (defined as alarms that are repeated 5 or more times per minute)	0
Stale alarms (defined as an alarm that remains in alarm condition for more than 24 hours).	0

Alarm Rationalization guidelines

It is recommended to rationalise the alarms, when the measured performance of an existing alarm system does not meet the requirements stated in S.No.8. During alarm rationalisation all the alarm tags/points shall be reviewed as per the procedure given in Sl. No 4. Station shall form an Alarm rationalisation/optimisation team at station consists of Two persons from operation with min 5 years of experience.

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- i. C&I Engineer of respective DCS for required changes in DCS and facilitate DCS data requirements.
- ii. EEMG to facilitate the process and to document results.
- iii. Electrical Engineer for electrical system alarms.
- iv. Maintenance planning

Other individuals with knowledge of the process of boiler, turbine and generator will be needed periodically.

Documents required for a thorough rationalization normally includes:

- Unit P&ID's
- Operating Instructions
- DCS configuration data
- Results from HAZOP or PHA (Potential sources for alarms are Hazards Analysis) reviews and
- Boiler, Turbine, and generator protection point lists, trip setpoints, and logic diagrams.
- Schedule of instruments, O&M manuals.
- Hardware scheme and wiring diagrams.

Access to the history to analog points via Process Book trends is particularly useful to select alarm setpoints that avoid nuisance alarms. Since the number of alarms to be rationalized is exceptionally large, alarm rationalization software is recommended for efficient use of workforce. The software should display all DCS alarm configuration details and allow for rapid documentation of results.

Guideline for Alarm rationalisation:

1. Alarm Priority Assignment

- a. In all cases, a documented consequence detrimental regarding safety, environment, equipment damage, or other cost shall constitute justification for every alarm (as described in section 4).
- b. "Return to Normal" alarms should be set to Journal priority. Alarms are to alert the operator when his action is required, not when conditions return to normal. However, return to normal can be important to identify in journals after an incident.
- c. "Bad I/O" alarms should be set to Low priority (or High if the loop has any Emergency priority alarm). This is an abnormal situation that requires operator

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response, but time duration is unknown.

- d. Actions by Operator (e.g., switch of loop to "Calibration" mode for work by the instrument tech) should be set to Journal priority. Many operator actions are journaled automatically by the DCS. Deviation from setpoint alarm is especially useful for those control loops where the setpoint is frequently moved.
 - e. Rate of Change alarms should be used sparingly.
2. Alarm Type Configuration Guidelines
 - a. For inputs: to avoid redundant alarms, BAD I/O alarm should only be configured on the raw input point.
 - b. For control loops: process alarm should be set on the controller point, not on the "raw" point.
 - c. If multiple "raw" points are being used as input to a function block then process alarms must be set on the raw points, and not on the composite point.
3. Grouped Alarms
 - a. Where Grouped Alarms are used, the individual tags in alarm will be displayed with their alarm indications shown on process graphics, but they will not appear on the Alarm Summary display. The Group Alarm for the group will appear on the Alarm Summary display.
 - b. If a Grouped Alarm for high vibration is activated by any one of 15 different instruments, the operator must be able to use his alarm shelving function on any of the 15 (making it unable to activate the common alarm), while the other 14 individual alarm points can still activate the common in the normal fashion.
4. Boiler, Turbine and Major Aux protection Alarms
 - a. Protection bypass: The entire protection bypass shall be of journal priority.
 - b. Local push buttons: All local push button pressed shall be in alarm. Control room push buttons activated shall be of journal priority as the operator operates the same.
5. Operator actions

All operator actions shall be for all of journal priority alarms.
6. Equipment Status and Auto Start/Stop

Equipment status shall be given Journal priority. Resultant changes to process conditions will be evident from temperatures, pressures, flows, etc. Switches used to automatically start or stop equipment shall be Journal priority when the automatic action is not abnormal (e.g., a pump automatically started/ stopped based upon header pressure low.)

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Annexure 6: List of Alarm references. (Reference)

Ref Of Alarm can be as follows (May Not limited to below mentioned):

1. Unit P&ID's.
2. Logic diagrams
3. Schedule of instruments.
4. O&M Manuals.
5. Operating instructions.
6. DCS/Panel configuration data like hardwire scheme, DCS internal alarms, Wiring diagram scheme.
7. Trip committee recommendation.
8. COS circulars/ OGNs/ODs/OINs
9. Change request forms.
10. Operating Instructions.
11. Results from HAZOP or PHA reviews.
12. System Alarms
 - a. Power supply alarms
 - i. UPS
 - ii. Charger
 1. Unit
 2. SAC
 - iii. Panel Power supply faults of
 1. DCS/RIO
 2. Flame scanners.
 3. VMS/TSI
 4. Interposing relay panels
 5. CJsCs
 6. Gravimetric feeders
 7. Generator instrument cabinet
 8. Profibus/Field bus cabinets
 9. EWLI
 10. ASLD
 11. Bunker Level
 12. Unit Level fire alarm system
 13. Network panels.
 14. Power distribution boards
 15. Power controller panel
 16. Overspeed protection panel
 17. LSR/LMU/Auto sync
 18. LP/HP bypass system
 19. Any MCB tripped.
 20. Local controllers
 - b. Network faults.
 - i. Switch fault.

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- ii. Port /WS fault
 - iii. Bus fault
 - iv. Time sync's out.
 - c. DCS Hardware faults
 - i. Controller fault
 - ii. I/O card faults
 - iii. Communication module fault
 - iv. RIO communication fault
 - v. Fan fault.
 - d. HMI
 - i. Workstation Fault
 - ii. Server Fault
 - iii. RAID array fault.
 - iv. Time sync
 - e. CER Temperature high
 - f. Instrument air dew point low.
- 13. Process alarms.
 - a. Closed Loop control Loop
 - i. Trip to manual
 - ii. Dev high
 - iii. Directional blocking
 - iv. Run up /run-down.
 - b. Measurements
 - i. TMS
 - 1. Deviation High
 - 2. Measurement faults
 - ii. DMS
 - 1. Deviation High
 - 2. Measurement faults
 - iii. Single measurements used in protection.
 - 1. Measurement fault
 - iv. Digital inputs
 - 1. Non-coincidence
 - c. Drives (OLCS)
 - i. Binary
 - 1. Tripped
 - 2. Fail to start.
 - 3. Fail to stop.
 - 4. Drive fault.
 - a. MCC Disturbance
 - b. Feedback trouble
 - c. Drive in local.
 - 5. Stand by not available.
 - 6. Drive maintenance.
 - 7. Out of sync.

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Annexure 7: Operators feedback questionnaire (Reference)

Questionnaires shall be used for obtaining the opinions of the operators regarding the usefulness of the alarms and collecting their observations and feedback. This could be useful in assessing the alarm system performance and reviewing the alarm system for carrying out improvements.

Operator questionnaires:

The main requirement of an alarm system is usability. As such, it is difficult to define and quantify the usability of an alarm system. However, an overall measurement of usability can be performed by a survey of operators' opinions using questionnaires. A typical questionnaire on the alarm system is shown below. However, the questions asked in survey questionnaires may vary from plant to plant, depending on plant processes, equipment, process control systems, and the alarm systems.

Introduction:

Dear Operator,

We are carrying out this survey to discover your views and opinions on the alarm systems in the plant and your valuable suggestions to improve the alarm systems. Through this survey we are looking for information that will help us to improve the performance of the alarm systems. Your inputs (particularly those on nuisance alarms, fleeting alarms, spurious alarms, details of incidents when faced alarm overloads, problems and limitations faced by you during abnormal/up-set plant conditions) will help us review our alarm strategies. Based on your feedback, we will review each of the relevant alarms and improve the overall performance of the alarm systems in the plant. Although the survey is being carried out by maintenance personnel, your views, opinions, and suggestions will be reviewed by a cross-functional team comprising of C&I, Electrical, Operation and EEMG. The inputs provided by you in the survey will remain confidential as we have tried to keep the questions in the survey objective in nature to avoid subjectivity.

You are requested to supply a response to each of the questions asked and submit the form online.

1. To which section/unit of the plant are your tasks related?
2. How long have you been using the present Alarm systems in the plant?
3. How do you rate the alarm system for providing you support during:
 - (a) Normal plant operations: Very Good / Good / Average / Poor / Very poor
 - (b) Emergency/Abnormal plant conditions: Very Good / Good / Average / Poor / Very poor

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4. What is your opinion about the number of alarms generated by the alarm system? Too many / Many but necessary / few but adequate / a few inadequate
5. How many alarms do you get per hour during the normal plant operation(Approx)?
6. How many alarms come up repeatedly within 5 minutes? (Can specify number.) Less than 25% / between 25% to 50% / between 50% to 75% / over 75%
7. Is each Alarm Message clear to understand?
All of them / most of them / some of them / few of them / very few
8. Do you know how to respond to each alarm?
All of them / most of them / some of them / few of them / very few
9. Do you get nuisance alarms? Often / sometimes / rarely
10. Are the alarms duplicated or redundant?
Most of them / some of them / few of them / very few / none
11. Nuisance alarms occur during plant shut down:
All of them / most of them / some of them / few of them / very few
12. Are there alarms that occur during a trip but are only relevant during normal plant operation? Most of them / some of them / few of them / very few / none
13. How many of the alarms require a positive action response from you? All of them / most of them / some of them / few of them / none
14. During a large plant trip or faults, how many alarms do you get...? In first 1 minute: ____
In 5 minutes: ____
In 15 minutes: ____
In 30 Minutes: ____
In an hour: ____
15. How many times do you look through the alarm display during a large plant trip/fault?
Several times in a minute / once a minute / once every 5 minutes / once in 10 minutes / keep alarm list in permanent display.
16. How often do alarms occur too fast for you during a large plant trip/fault? Always / quite often / sometimes / few times / never

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17. How many times during a large plant trip/fault do you accept/acknowledge the alarms without reading and understanding them?
Always / quite often / sometimes / few times / never
18. During a large plant trip/fault, how many safety related alarms can you identify? All of them / most of them / some of them / few of them / none
19. The procedures for getting changes made in alarms, alarm settings are: Restricted & cumbersome / strict & safe / easy & cautious / lenient.
20. Do the site C&I engineers put efforts into improving the alarm systems? Yes, too much / as and when asked for / too little / not sufficient.
21. What features of the alarm systems do you not like?
22. What main features of the alarm system do you like most?
23. What additional features in the alarm system would help you run the plant better?
24. What features in the alarm system do you find are of no use to you?
25. Any other comments/suggestions